

MYOUNGCHUL KIM

Data Scientist

English: Fluent

Japanese: Fluent, JLPT N1

Korean: Native

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Greater Tokyo Area, Japan

MyoungchulK

myoungchulkim

ABOUT ME

Data Scientist with a strong foundation in **AstroParticle Physics**, statistical modeling, and large-scale data analysis. Experienced in **LLM**, **RAG**, **LangGraph**, multi-agent systems, **SFT**, **RLHF**, **ML pipeline automation**, and **Python-based ML Engineering**. Proven track record across academic research and industry AI applications.

TECHNICAL SKILLS

Coding Tools:

PythonC++Git / GitHubVimJupyterLatexROOTHTCondorCVMFS

Data Analytics:

NumPySciPyScikit-learnPandasGeoPandasSQLMatplotlibSeaborn

Modelling:

TensorFlowPyTorchUnsupervised LearningCNNGNNTime seriesEnsemble MethodsStatsmodelsLLM AgentRAGRLHFSFTHugging Face

Deployment:

AWSGCPDockerDatabricksDagsterLangChainLangGraphFastAPIStreamlitRaspberry Pi

EXPERIENCE

Data Scientist

Fracta

November 2024 – July 2026

Tokyo, Japan

AI-Driven Infrastructure Assessments

November 2024 – July 2026

Role: Data Scientist

Team size: 4

- Developed an **LLM-powered multi-agent system** leveraging **LangGraph** and **RAG** for automated raw data preprocessing prior to **ML** ingestion.
- Deployed **gradient-boosted models** – with **SHAP** and **survival analysis** – to predict **Likelihood of Failure (LOF)** across large-scale networks and deliver client-facing risk assessments.
- Engineered an end-to-end **MLOps** pipeline in **Dagster**, deployed via **Docker** on **AWS EKS** with **CI/CD**, reducing manual intervention at scale.
- Improved a **Bayesian (Stan; variational inference + MCMC)** imputer and built a **GNN (PyTorch Geometric)** counterpart to recover missing attributes, with **uncertainty quantification**.

Data Science Instructor

Le Wagon

October 2024 – Ongoing

Tokyo, Japan

Data Science & AI Bootcamp Instruction

October 2024 – Ongoing

Role: Teaching Assistant

Team size: 13

- Instructed **Python** for Data Science, covering data extraction, manipulation, visualization, and **statistics**.
- Guided students through **ML/DL** workflows using **Scikit-Learn** and **neural network** architecture design.
- Delivered training in **ML Engineering**, including **Python** package deployment on **GCP** and ethical implications of **AI**.

Python Data Scientist/Analyst

Turing

📅 July 2024 – January 2025

📍 Remote

Multilingual SFT and RLHF Implementation for Next-Generation AI

📅 July 2024 – January 2025

👤 Role: LLM Dataset Creation

👥 Team size: 6

- Performed **Supervised Fine-Tuning (SFT)** and **Reinforcement Learning with Human Feedback (RLHF)** across **English**, **Korean**, and **Japanese** datasets to improve **LLM** alignment.
- Analyzed model performance via **Loss Pattern** tracking and **Output-Based Evaluation** to enhance robustness and reliability.
- Curated model responses through **Comparative Analysis** and **Preference Pair** evaluation, ensuring contextually accurate outputs.

AI Training for Japanese Writers

Outlier

📅 July 2024 – October 2024

📍 Remote

Fine-Tuning and Validation of AI-Generated Japanese Voice

📅 July 2024 – October 2024

👤 Role: LLM evaluation

👥 Team size: 12

- Validated **AI-generated voice** outputs, assessing quality via **data augmentation**, **pitch**, and **speech rate** for natural and contextually accurate results.
- Applied **linguistic expertise** in Japanese to evaluate **voice model** performance against clarity, consistency, and **cultural accuracy** benchmarks.

Graduate Research Assistant, Ph.D.

International Center for Hadron Astrophysics (ICEHAP)

📅 April 2017 – December 2023

📍 Chiba University, Japan

Search for Ultra-high Energy Neutrinos from Askaryan Radio Array (ARA) by Template Method 🔗

📅 April 2018 – December 2023

👤 Role: Project Leader

👥 Team size: 6

- Classified astronomical signals using **PCA** on 200 TB of **radio-frequency** data, optimized via **Frequentist Statistics** and **Pseudo Experiments**.
- Implemented **FFT**, **Interferometry**, and **Matched Filter** for feature extraction on large-scale **CPU & GPU** clusters using scientific **Python** and **C++**.
- Evaluated results via **statistical significance**, **systematic uncertainty**, and **Monte Carlo simulation**; pipeline adopted by international collaborators.

Development of In Situ Antenna Model for Simulation

📅 April 2017 – March 2019

👤 Role: Data Management

👥 Team size: 4

- Performed high-precision calibration for **radio-frequency antennas** to measure and characterize physical properties.
- Extracted feature patterns from raw data through **noise & signal analysis**, contributing to simulation model development.

Graduate Research Assistant, MSc.

Neutrino AstroParticle Physics Lab (NAPPL)

📅 March 2015 – February 2017

📍 SungKyunKwan University, South Korea

IceCube Camera System to Study Properties of the Antarctic Ice 🔗

📅 March 2015 – February 2017

👤 Role: Project Leader

👥 Team size: 5

- Classified intrinsic camera noise under extreme low-temperature conditions using **feature extraction** and **image analysis**.
- Developed **Python** package for camera control via **Raspberry Pi**, enabling automated data collection.

- Evaluated camera system performance by applying **statistical techniques** to large-scale **image datasets**.

PROJECT

AI Music Generation via RNN

Le Wagon Data Science & AI Bootcamp

 January 2024 – March 2024

 Le Wagon, Tokyo

 Role: Data Management

 Team size: 4

- Architected **RNN** to learn musical patterns from large classical music datasets encoded in numerical form, enabling customizable music generation.
- Deployed end-to-end pipeline via **FastAPI** and **Streamlit**, integrating generated outputs with **Ableton**.
- Managed data preprocessing and model training workflows, ensuring scalable and reproducible **deep learning** pipelines.

EDUCATION

Data Science & AI Bootcamp

Le Wagon

 January 2024 – March 2024

 Japan

- Studied **Python** for Data Science, covering **data extraction**, **manipulation**, **visualization**, **statistics**, and **linear algebra**.
- Built end-to-end **ML/DL** workflows using **Scikit-Learn** and designed **neural network** architectures for practical applications.
- Developed **Python** packages for large-scale data tasks on **GCP**, with exposure to **ML Engineering** and **AI ethics**.

Completion of Ph.D. program (ABD), AstroParticle Physics

Chiba University

 April 2017 – December 2023

 Japan

- Research topic: Search for Ultra-high Energy Neutrinos Using Eight Years of Data from Two ARA Stations by the Neutrino Template Method

Master of Science, AstroParticle Physics

SungKyunKwan University

 March 2015 – February 2017

 South Korea

- Thesis title: Performance study of camera system for the IceCube-Gen2 detector

Bachelor in Science, Physics

SungKyunKwan University

 March 2011 – February 2015

 South Korea

SELECTED PUBLICATIONS

Journal Articles

- P. Dasgupta, M. S. Muzio, *et al.*, “Progress Towards a Diffuse Neutrino Search in the Full Livetime of the Askaryan Radio Array,” *PoS*, vol. ICRC, p. 1226, 2023.
- M. Kim *et al.*, “Enhanced Ultra-High Energy Neutrino Search at the Askaryan Radio Array using Template-based Techniques,” *PoS*, vol. ICRC, p. 1148, 2023.
- D. Bose, M. Jeong, K. Woosik, J. Kim, M. Kim, C. Rott, *et al.*, “PINGU camera,” *PoS*, vol. ICRC, p. 1145, 2015.

AWARDS



Japanese Government Monbukagakusho Scholarship (MEXT)

Graduate Research Assistant in Ph.D., 2017 – 2020



Teaching Assistant (T.A.) of Korea & Japan Joint Government Scholarship

Teaching Assistant for a freshman Korean students, 2017 – 2018



BK21+ Research student scholarship

Graduate Research Assistant in Msc., 2015 – 2017



Operating Assistant scholarship

Physics Experiment Assistant, 2016 – 2017



CK Research student scholarship

Research Assistant in Bsc., 2014

INTERESTS

Classical Music

Orchestra

Contrabass

Universe

Fourier Transform